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COMPARISON AND RELIABILITY OF THREE GONIOMETRIC METHODS FOR MEASURING FOREARM SUPINATION AND PRONATION[1,2]

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Summary.—Active supination and pronation range of motion of the right forearm by 10 men and 21 women were measured with a goniometer using three methods. For Method One the moveable arm was aligned with a pencil held in the hand; for Method Two the moveable arm was placed over the anterior and posterior surface of the distal forearm for supination and pronation, respectively; and for Method Three the moveable arm was aligned parallel with a visualized line connecting the distal ulna and radius. Supination range of motion was greatest for Method One, followed by Methods Three and Two, respectively. Pronation range for Method One was less than for the two other methods. Intraclass correlation coefficients ranged from .81 to .97 for three trials within the test session and from .86 to .96 between the test and a 1-hr. retest session for both actions using all three methods. Method One was the most reliable but appeared invalid for measuring supination because of movements of the 4th and 5th metacarpal bones. Methods Two and Three appeared appropriate for clinical application but not interchangeable.

Therapeutic interventions designed to increase restricted supination and pronation range of motion are used routinely in physical rehabilitation following conditions such as trauma, surgical repairs, and immobilization. Valid and reliable methods for measuring supination and pronation are important for documenting movement limitations and changes that result from treatment interventions. Of the goniometric methods available for measuring the two actions, two methods are documented in basic textbooks (Norkin & White, 1985; Clarkson & Gilewich, 1989). For the first method the axis of the goniometer is over the head of the third metacarpal bone and the moveable arm is aligned with a pencil held in the patient's fisted hand (Clarkson & Gilewich, 1989). For the second method the axis of the goniometer is medial to the ulnar styloid process for supination and lateral to the ulnar

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[2]A detailed account is on file in Document APD2001-015 with the Archive for Psychological Data, P.O. Box 7922, Missoula, MT 59807-7922. Remit $15.00 for photocopy.

styloid process for pronation: the moveable arm is put across the anterior and posterior aspect of the distal forearm, respectively (Norkin & White, 1985). The present study included a third method that aligned the moveable arm with a visualized line connecting the distal aspects of the ulna and radius. The purposes of the study were to compare the maximal active supination and pronation range of motion among the three methods and to examine their within-test and test-retest intrarater reliability.

Thirty-one nondisabled subjects (10 men and 21 women) between 19 and 40 years of age participated with informed consent. An 8-in., 180° stainless steel goniometer with 1° increments was used for all measurements. Each subject was seated on a high-low table with the right arm adducted to the trunk, the elbow flexed to 90°, and the forearm in a neutral position with the thumb pointing toward the ceiling. One examiner aligned the goniometer for each of the three methods. When the end point of motion was reached, a second examiner read and recorded the measurements. Three trials of each method were collected during test and retest sessions separated by 1 hr.

Method One: For both supination and pronation the subject held a pencil that was placed over the distal palmar groove of the hand. The axis of the goniometer was over the head of the third metacarpal with the stationary arm held parallel to the midline of the humerus and the moveable arm held parallel to the pencil (Clarkson & Gilewich, 1989).

Method Two: For supination, the hand was fisted and the axis of the goniometer was medial to the ulnar styloid process with the stationary arm parallel to the midline of the humerus. The moveable arm was placed across the anterior surface of the distal forearm. For pronation, the axis of the goniometer was lateral to the ulnar styloid process and the stationary arm was held parallel to the midline of the humerus. The moveable arm was placed across the posterior surface of the distal forearm (Norkin & White, 1985).

Method Three: For this alternative method the hand was fisted, and the examiner placed the axis of the goniometer over the head of the third metacarpal. The stationary arm was held parallel to the midline of the humerus and the moveable arm was held parallel to a visualized line connecting the styloid processes of the ulna and the radius.

Analysis of variance indicated a difference among the three methods for both supination and pronation ($p \leq .001$). For supination, *post hoc* tests indicated differences between all pairwise comparisons ($p \leq .002$): Method One had the greatest supination measurement (112°, $SD = 16°$), followed by Method Three (100°, $SD = 11°$) and Method Two (95°, $SD = 7°$). Method One supination was confounded by observed movements of the fourth and fifth metacarpal bones. Pronation range of motion using Method One (69°, $SD = 13°$) was less than that for Method Two (72°, $SD = 11°$) and Method Three

(73°, $SD = 11°$, $p < .007$), but pronation using Methods Two and Three did not differ.

The intraclass correlation coefficients (ICC, Model 2) within the test session indicated good intratester reliability for supination and pronation for each of the three methods (ICC range: .81–.97). Reliability between the tests and retests was also good for supination and pronation for all three methods (ICC, Model 2, range: .86–.96).

Differences in the range of motion among the three methods indicated that they should not be used interchangeably to measure forearm supination. Method One appeared to overestimate supination range of motion compared to Methods Two and Three, probably because movement of the fourth and fifth metacarpal bones added to the supination range of motion. Methods Two and Three yielded active range of motion measurements of supination and pronation with good intratester reliability, and both methods appeared valid and appropriate for clinical use. Although pronation did not differ between these two methods, differences in their supination measurements suggested that they should not be used interchangeable to measure supination range of motion.

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